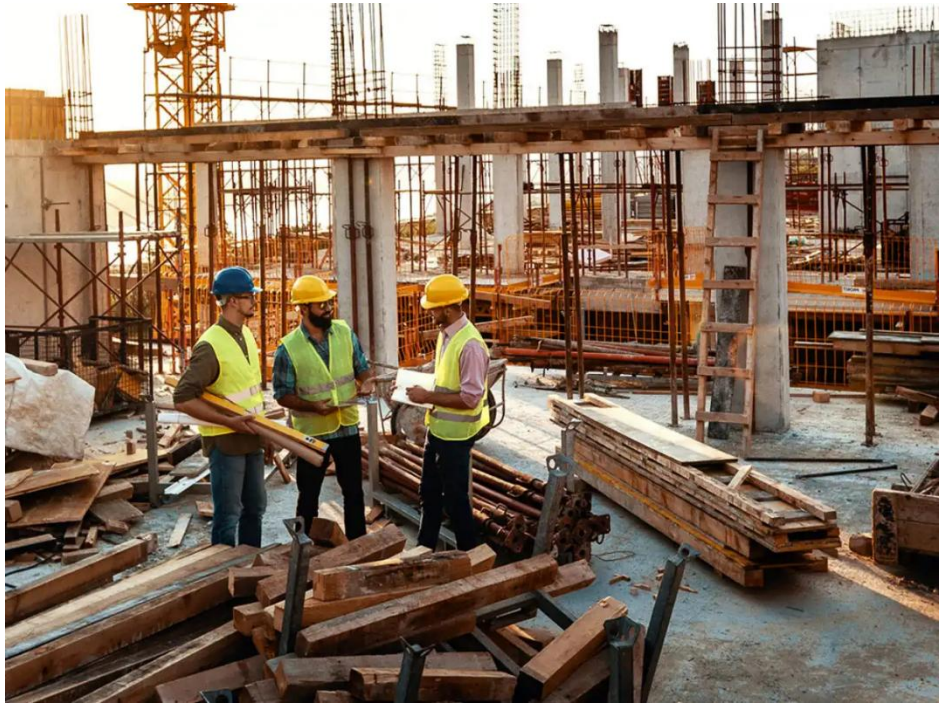


MACHINE LEARNING BASED SMART HELMET FOR PERSONALIZED CONSTRUCTION SAFETY MONITORING

Project ID: R26-SE-042

RESEARCH PROBLEM

RESEARCH PROBLEM(Cont.)

Users & Stakeholders



Users	Stakeholders
Construction Workers	Construction Companies / Management
Site Supervisors / Safety Officers	Regulatory Authorities (Safety Compliance)
	Insurance Providers
	System Administrators



Benefits of Addressing the Problem

- Improved worker safety and reduced accidents
- Early detection of hazardous conditions
- Increased trust between workers and management
- Better compliance with safety regulations
- Reduced financial losses (delays, legal costs)
- Enhanced productivity and efficiency

RESEARCH PROBLEM(Cont.)

Existing Solutions & Their Limitations

System / Study	Environmental Monitoring	Physiological Monitoring	Machine Learning	Real-time Alerts	Personalization	Integrated Risk Analysis
IoT Smart Helmet (2025)	✓	×	×	✓	×	×
BLE-LoRa Smart Helmet	✓	×	×	✓	×	×
Smart Helmet and Jacket System	✓	✓	×	✓	×	×
Coal Miner Wearable Device	✓	✓	×	✓	×	×
SafeHelmet-AI	✓	✓	✓	✓	×	Partial
AI Deep Learning Helmet	✓	✓	✓	✓	×	×
Smart Wear Guardian	✓	✓	✓	✓	×	×
Personalized Wearable Sensors Study	✓	✓	✓	×	✓	×
Thermal Comfort Wearable Study	✓	✓	✓	×	✓	×
Proposed System	✓	✓	✓	✓	✓	✓



Key Limitations

- Reliance on fixed threshold-based alerts
- Lack of personalization for individual workers
- Environmental and physiological data analyzed separately
- Limited real-time adaptability
- Mostly prototype-level (limited real-world deployment)

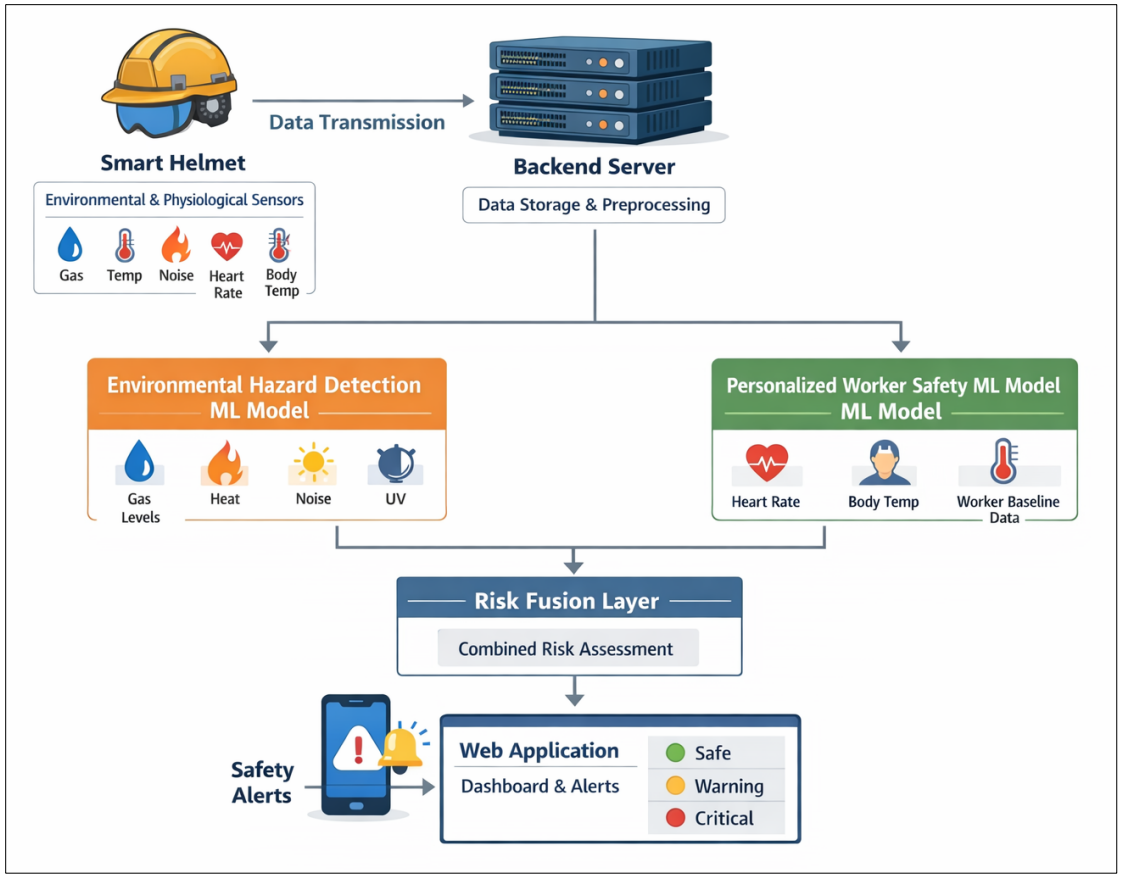
SOLUTION OVERVIEW

Proposed Solution:

Machine Learning–Based Smart Helmet System

Key Features:

- Real-time environmental & physiological monitoring
- Dual machine learning models
- Personalized safety analysis
- Integrated risk assessment (risk fusion)
- Adaptive, real-time alerts



Conceptual Diagram

How does this address the gap:

- Moves beyond threshold-based detection
- Combines multiple data sources
- Provides personalized safety monitoring
- Enables context-aware decision making

Achieved by:

Multi-Modal Safety Intelligence Component



IT21200892 |METHYA KARIYAPPERUMA

Multi-Modal Safety Intelligence Component

MULTI-MODAL SAFETY INTELLIGENCE COMPONENT

Novel Contributions of the Proposed Work

- Integration of environmental and physiological data for unified safety analysis
- Dual machine learning approach for hazard detection and worker condition analysis
- Personalized safety monitoring based on individual baseline data
- Risk fusion mechanism to combine multiple risk factors into a single safety score
- Adaptive alerting instead of fixed threshold-based detection



MULTI-MODAL SAFETY INTELLIGENCE COMPONENT

Objective

To design, develop, and evaluate a machine learning–based smart helmet system that integrates environmental hazard detection and personalized worker safety monitoring to generate real-time safety risk predictions with at least 85% accuracy under simulated construction conditions.

Sub-Objective

- To design and implement a smart helmet prototype equipped with environmental and physiological sensors capable of collecting real-time data.
- To develop and train two machine learning models for environmental hazard detection and personalized worker safety analysis.
- To design and implement a backend system and web-based dashboard for real-time data processing, visualization, and automated safety alert generation.
- To evaluate the performance of the proposed system in terms of prediction accuracy, response time, and reliability using experimental testing scenarios.



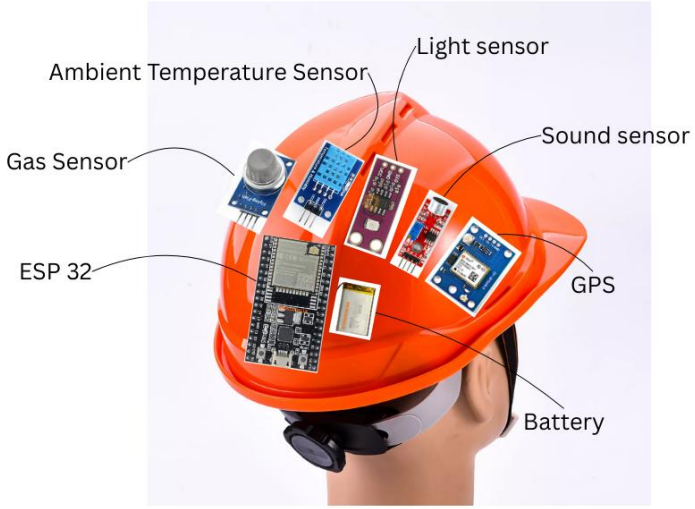
MULTI-MODAL SAFETY INTELLIGENCE COMPONENT

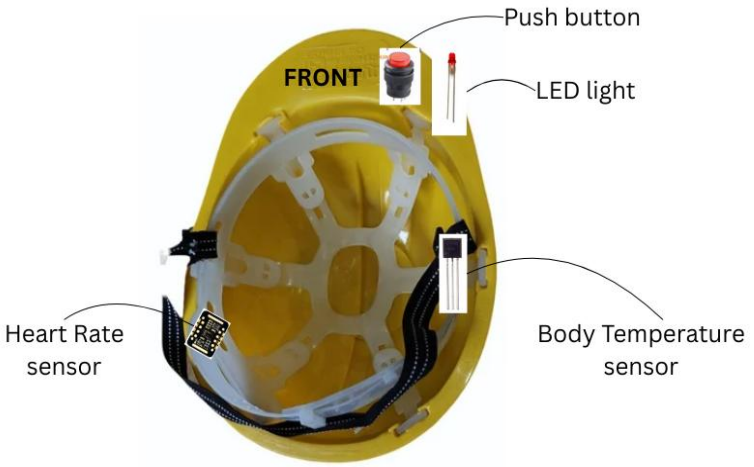
Application of Domain Knowledge

Research Approach
- Design Science Research (DSR)
- Experimental evaluation

Technologies Used
- IoT-based wearable sensors (ESP32, gas, temperature, HR sensors)
- Machine Learning (Random Forest / Gradient Boosting)
- Edge & Cloud Processing
- Web Technologies (React, Node.js, APIs)

Key Technical Methods
- Real-time data collection & preprocessing
- Feature extraction (baseline deviation, trends)
- Dual-model ML analysis
- Risk fusion for final safety prediction





MULTI-MODAL SAFETY INTELLIGENCE COMPONENT

Validation & Evaluation

- Accuracy, Precision, Recall, F1-score
- Cross-validation & test datasets
- Real-time response evaluation

Security

- Secure data transmission (HTTPS)
- Authentication & access control
- Data anonymization

Data Availability

- Collected via smart helmet sensors
- Environmental + physiological data
- Controlled experimental scenarios

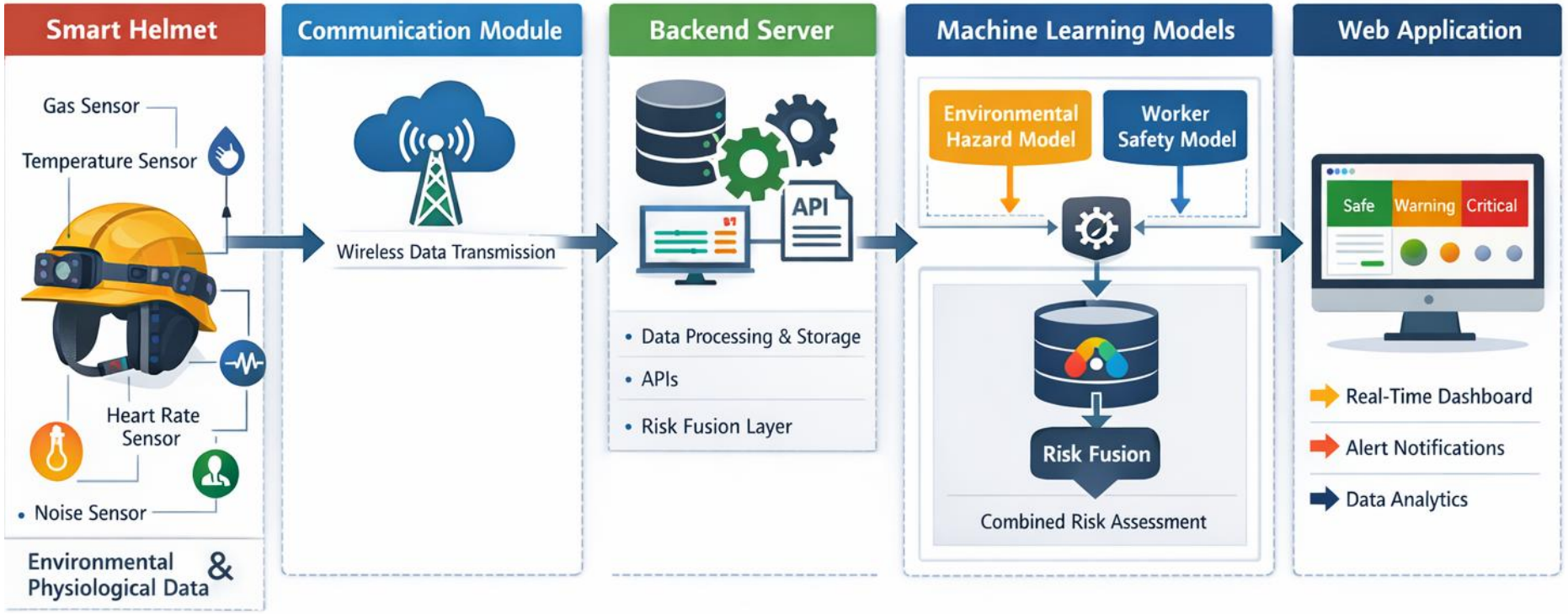
Ethical Considerations

- Informed consent from participants
- No misuse of personal data
- Compliance with research ethics guidelines



MULTI-MODAL SAFETY INTELLIGENCE COMPONENT

System Overview



MULTI-MODAL SAFETY INTELLIGENCE COMPONENT

Investment & Cost

- Hardware: ~12,000 LKR per unit
- Software: Open-source (low cost)
- Cloud: Subscription-based

Target Market & Value

•Construction companies & safety managers
•Improves safety, reduces accidents
•Builds trust and ensures compliance

Revenue & Pricing

•Device sales (one-time)
•Monthly subscription (dashboard & analytics)
•Maintenance services

Intellectual Property

•ML models & system design
•Potential for patent and software protection



MULTI-MODAL SAFETY INTELLIGENCE COMPONENT

Work Breakdown Structure



MULTI-MODAL SAFETY INTELLIGENCE COMPONENT

Gantt Chart





Thank You